CS - 4031 Compiler Construction

Assignment # 01

Automata Design

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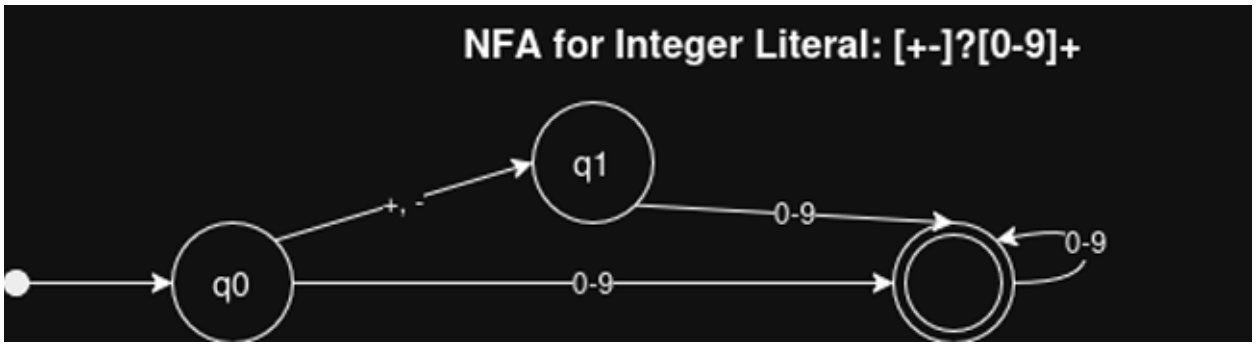
1. Regular Expressions & NFA Diagrams

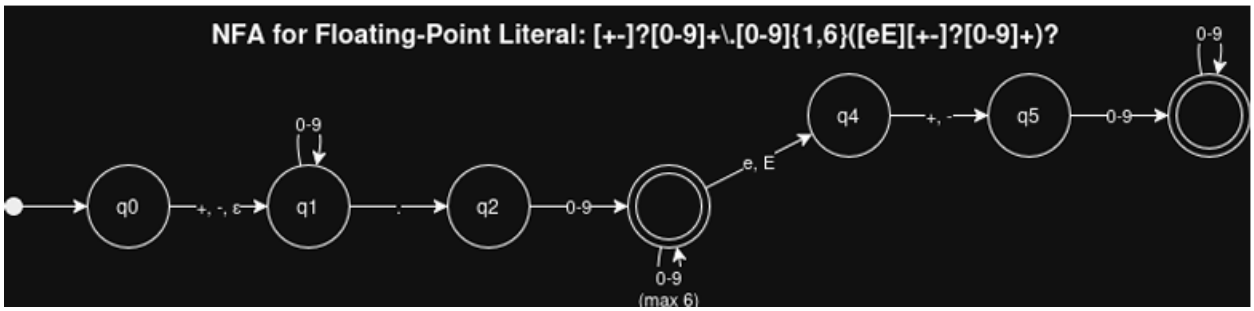
1.1 Integer Literal

Regex: [+-]?[0-9]+



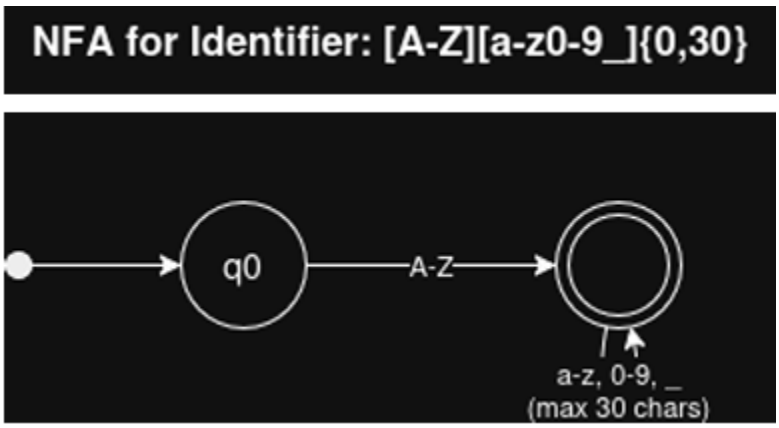
1.2 Floating-Point Literal

Regex: [+-]?[0-9]+\.[0-9]{1,6}([eE][+-]?[0-9]+)?



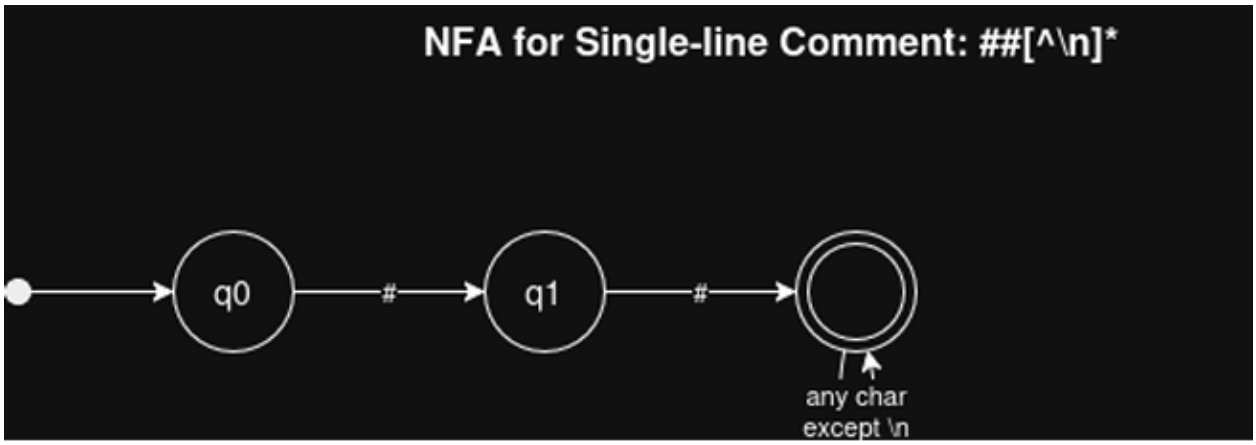
1.3 Identifier

Regex: [A-Z][a-z0-9_]{0,30}



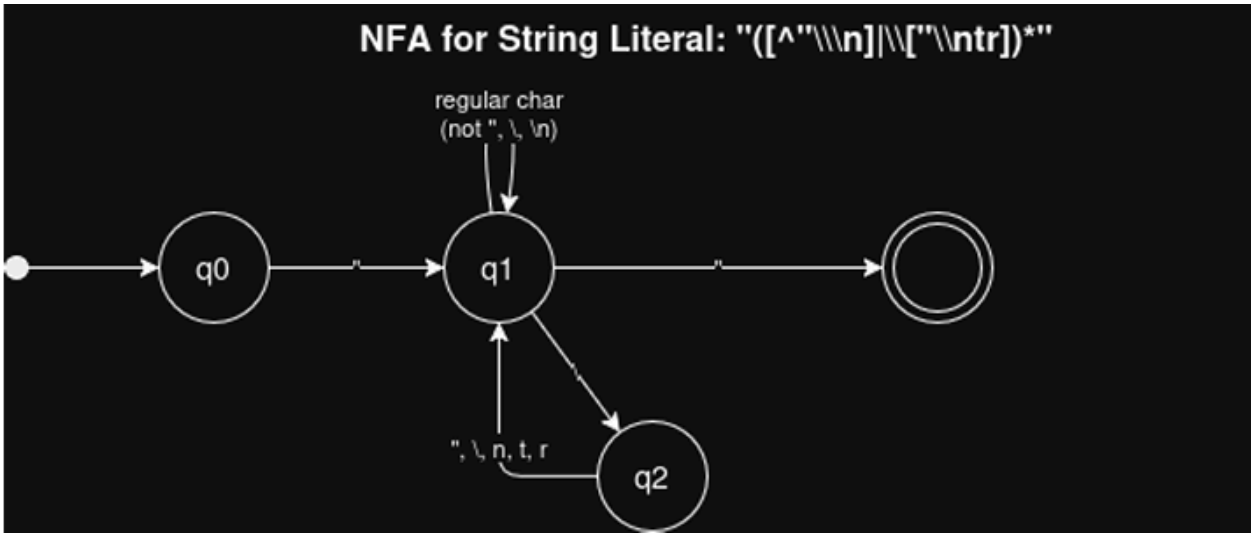
1.4 Single-line Comment

Regex: ##[^\n]*



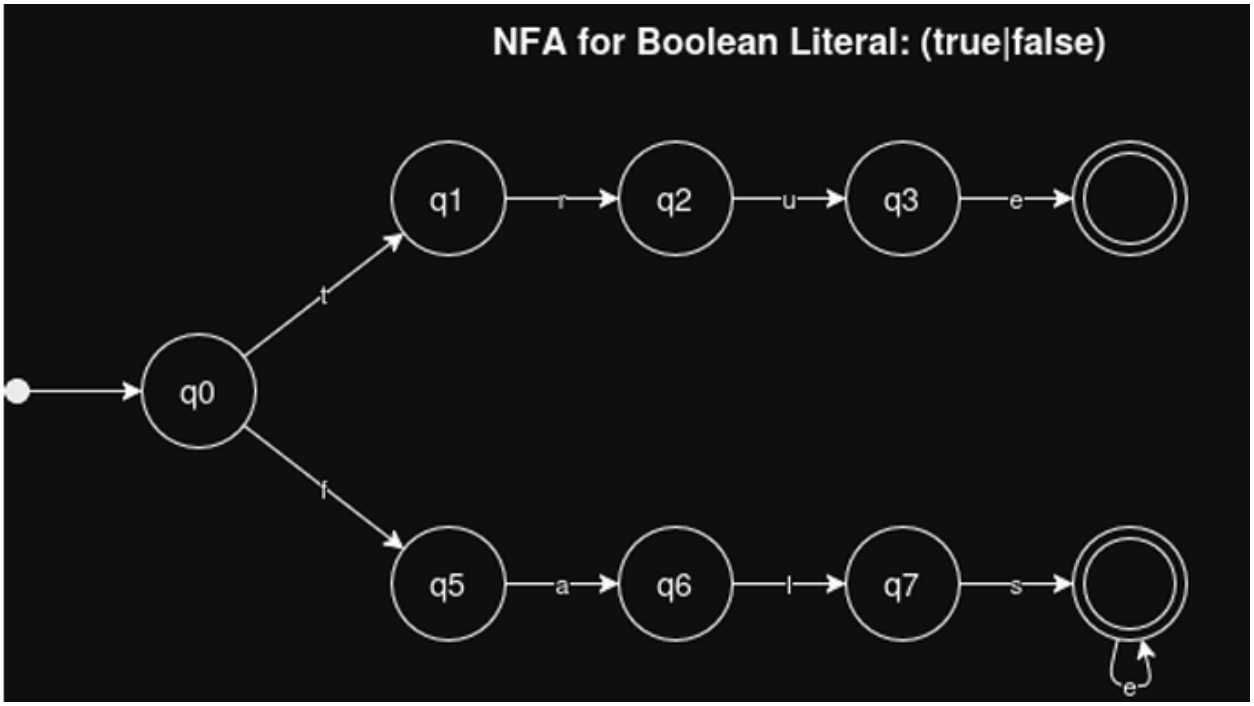
1.5 String Literal

Regex: "([^"\\\n]|\\["\\ntr])*"



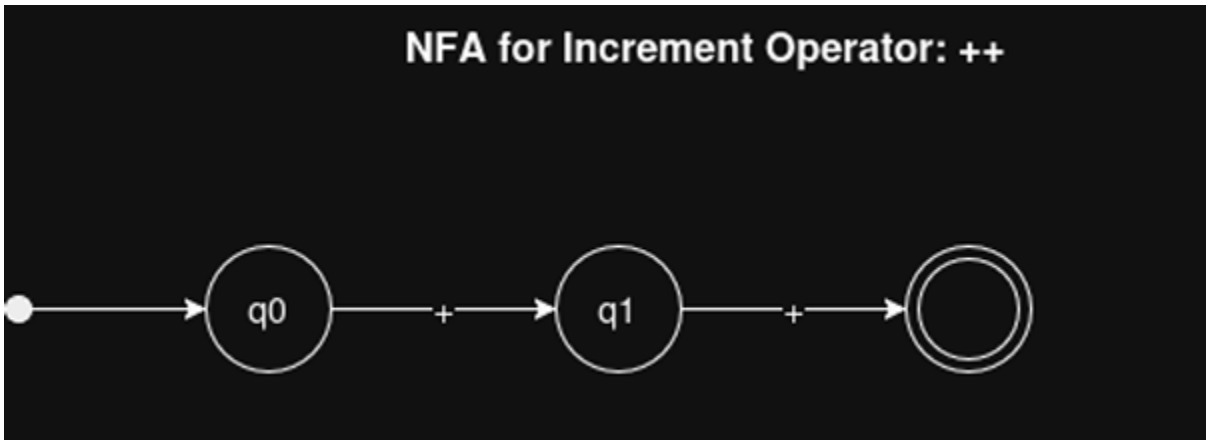
1.6 Boolean Literal

Regex: (true|false)



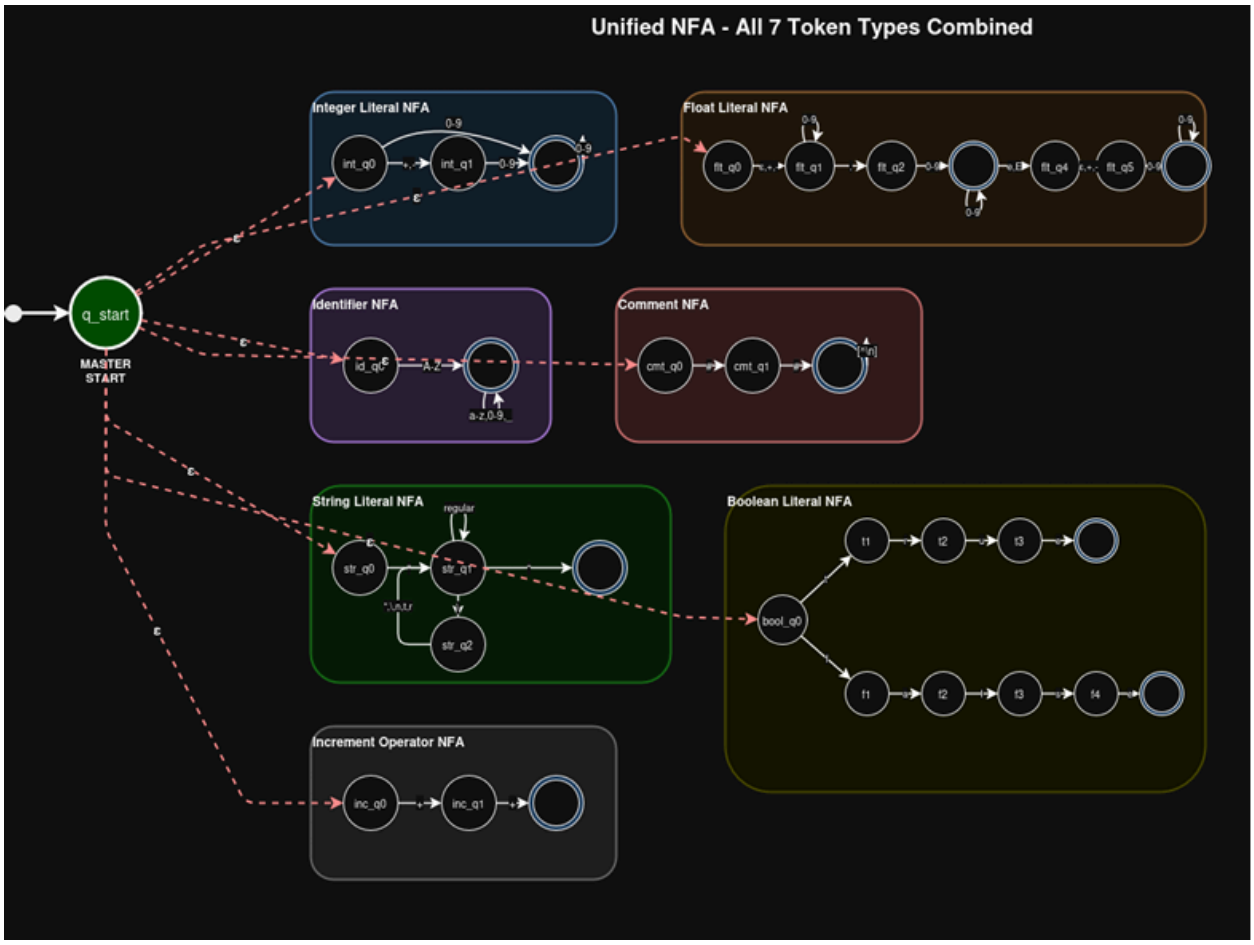
1.7 Increment Operator

Regex: \+\+

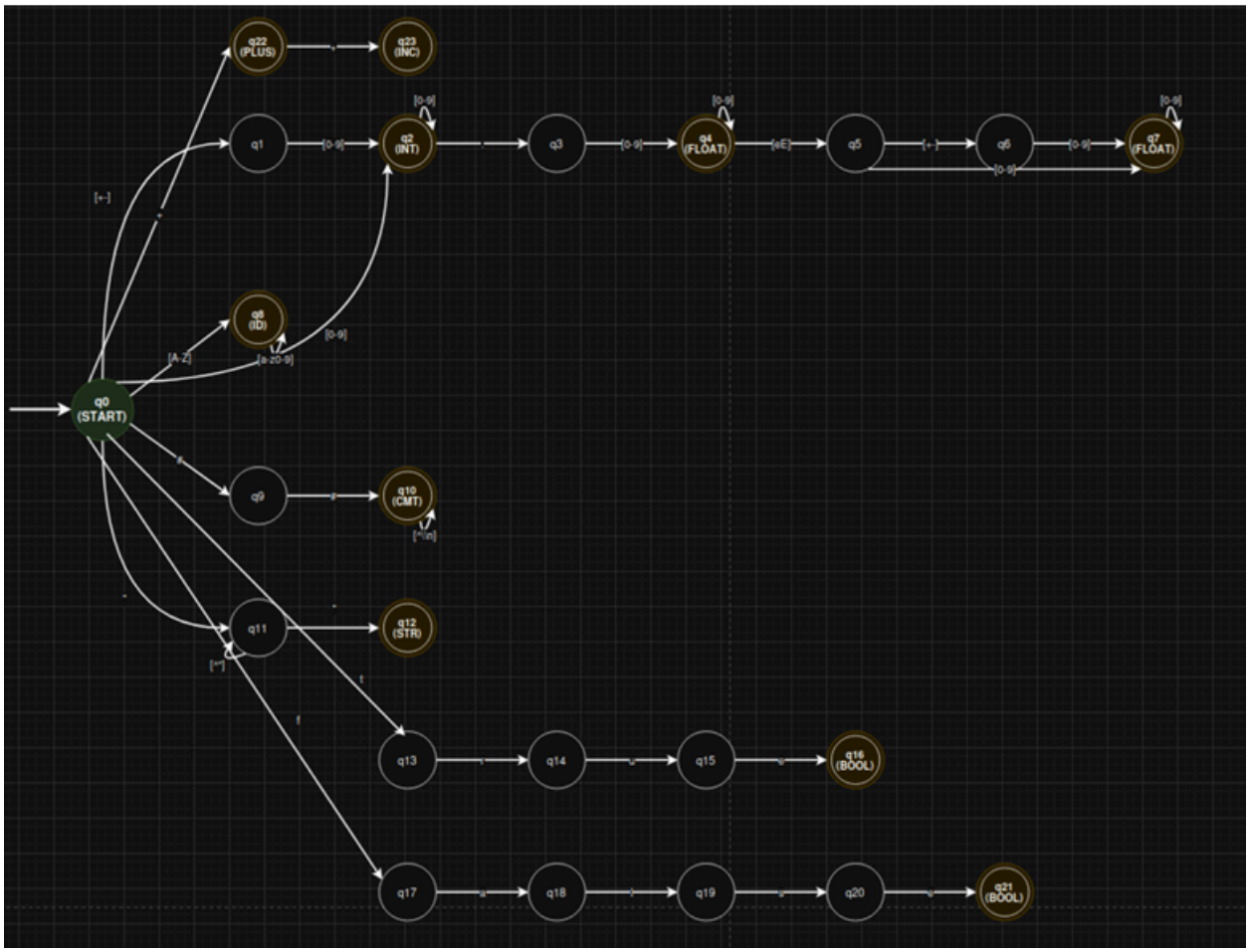


2. Combined Automata

2.1 Unified NFA



2.2 Minimized DFA



3. Transition State Tables (Minimized DFA)

3.1 Integer Token

State	[0-9]	+ / -	Other	Type
q0	q2	q1	-	Start

q1	q2	-	-	-
q2	q2	-	Accept	INTEGER

3.2 Float Token

State	[0-9]	+ / -	.	e / E	Other	Type
q0	q2	q1	-	-	-	Start
q1	q2	-	-	-	-	-
q2	q2	-	q3	q5	Accept	INTEGER
q3	q4	-	-	-	-	-
q4	q4	-	-	q5	Accept	FLOAT
q5	q7	q6	-	-	-	-
q6	q7	-	-	-	-	-
q7	q7	-	-	-	Accept	FLOAT

3.3 Identifier Token

State	[A-Z]	[a-z]	[0-9]	Other	Type
q0	q8	-	-	-	Start
q8	-	q8	q8	Accept	IDENTIFIER

3.4 Comment Token

State	#	[^ \n]	\n	Other	Type
q0	q9	-	-	-	Start
q9	q10	-	-	-	-
q10	q10	q10	Accept	q10	COMMENT

3.5 String Token

State	"	[^ \n "]	\	Other	Type
q0	q11	-	-	-	Start

q11	q12	q11	q11	-	-
q12	-	-	-	Accept	STRING

3.6 Boolean Token

State	t	f	r	u	e	a	l	s	Type
q0	q13	q17	-	-	-	-	-	-	Start
q13	-	-	q14	-	-	-	-	-	-
q14	-	-	-	q15	-	-	-	-	-
q15	-	-	-	-	q16	-	-	-	BOOLEAN
q17	-	-	-	-	-	q18	-	-	-
q18	-	-	-	-	-	-	q19	-	-
q19	-	-	-	-	-	-	-	q20	-

q20	**-**	**-**	**-**	**-**	**q21**	**-**	**-**	**-**	**BOOLEAN**

3.7 Whitespace Token

State	**Space / Tab / Newline**	**Other**	**Type**
q0	**q23**	**-**	**Start**
q23	**q23**	**Accept**	**WS**